

Sina Sajadmanesh

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Education

Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering	May 2019 – Aug 2023
Sharif University of Technology, MSc in Information Technology Engineering	Sept 2014 – Sept 2016
University of Isfahan, BSc in Computer Software Engineering	Sept 2009 – Feb 2014

Experience

AI Engineer, Sony AI – Zurich, Switzerland Oct 2023 – present

- Working on privacy-preserving vision foundation models
- Implemented easy-to-use APIs for vision foundation model inference and deployment
- Developed a comprehensive multi-task learning framework based on OpenMMLab libraries and PyTorch, supporting various vision tasks, new model architectures, and mixed-precision and distributed training
- Implemented an end-to-end open-world classification pipeline using MMClassification, HuggingFace transformers, and CLIP
- Improved model training and inference speed by optimizing the data pipeline, using mixed-precision training, and leveraging distributed training on multiple GPUs

Research Assistant, Idiap Research Institute – Martigny, Switzerland May 2019 – Aug 2023

- Worked as a doctoral researcher on differentially private machine learning with graph neural networks
- **3 published papers** in top-tier conferences (CCS, USENIX Security, and NDSS) with **185+ citations**
- **7 invited talks** at top universities and research institutions, including Imperial College, UIC, and Twitter
- **6 open-source projects** with **110+ stars** on GitHub
- **1 short course** taught on "Trustworthy Machine Learning" at [Artificial Intelligence Doctoral Academy](#)
- **Finalist** in [CSAW Applied Research Competition](#) for the best paper award in computer security in Europe

Visiting Collaborator, The Alan Turing Institute – London, UK Mar 2023 – Mar 2023

- Co-organized a workshop on "[Privacy and Fairness in AI for Health](#)" with 60+ attendees

Research Intern, Brave Software – Remote Mar 2022 – May 2022

- Developed a novel privacy-preserving **federated learning** framework for **neural bandit models** under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite

Research Assistant, Sharif University of Technology – Tehran, Iran Nov 2014 – May 2019

- Worked on various research projects including privacy-preserving machine learning, web data science, and social and information network analysis.
- **4 published papers** in top-tier venues (WWW, TKDD, IoTJ, ASONAM) with **430+ citations**
- **5+ press releases** in top-tier media outlets, including [MIT Technology Review](#), [France 24](#), and [The Independent](#)
- **2 open-source projects** with **20+ stars** on GitHub
- **1 semester course** taught on "Fundamentals of Programming with Python" at Sharif University of Technology

Skills and Expertise

Programming Languages: Python, C++, Java, SQL, Shell, \LaTeX

Machine Learning and AI: PyTorch, TensorFlow, HuggingFace, OpenMMLab, PyTorch-Lightning

MLOps and DevOps: Weights & Biases, Docker, GitHub, Dask, Neptune, Linux

Privacy-Enhancing Technologies: Flower, Opacus, Auto-DP

Community and Professional Service

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), Graph Neural Networks User Group Meetup (2021), Twitter Machine Learning Seminar (2021)

Organizing Committee: Privacy and Fairness in AI for Health (2023)

Program Committee: PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2024), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

Publications

Argus: A Compact and Versatile Foundation Model for Vision 2025-6

Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al.
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees Mar 2024

Sina Sajadmanesh, Daniel Gatica-Perez
ACM International Conference on Web Search and Data Mining (WSDM)

Privacy-Preserving Machine Learning on Graphs Aug 2023

Sina Sajadmanesh
Doctoral Thesis - Swiss Federal Institute of Technology (EPFL)

GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation Aug 2023

Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, et al.
USENIX Security Symposium (USENIX Security)

Locally Private Graph Neural Networks Nov 2021

Sina Sajadmanesh, Daniel Gatica-Perez
ACM Conference on Computer and Communications Security (CCS)

When Differential Privacy Meets Graph Neural Networks June 2020

Sina Sajadmanesh, Daniel Gatica-Perez
Technical Report, ArXiv e-prints

A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics May 2020

Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al.
IEEE Internet of Things Journal (IoTJ)

Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks Aug 2019

Sina Sajadmanesh, Sogol Bazargani, Jiawei Zhang, Hamid R. Rabiee
ACM Transactions on Knowledge Discovery from Data (TKDD)

NPGLM: A Non-Parametric Method for Temporal Link Prediction June 2017

Sina Sajadmanesh, Jiawei Zhang, Hamid R. Rabiee
Technical Report, ArXiv e-prints

Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web Apr 2017

Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al.
International World Wide Web Conference Companion (WWW)

Predicting Anchor Links between Heterogeneous Social Networks Aug 2016

Sina Sajadmanesh, Hamid R. Rabiee, Ali Khodadadi
International Conference on Advances in Social Networks Analysis and Mining (ASONAM)